Customer Churn Analysis Report

Churn Rate, Customer Lifetime Value (CLV), MRR Loss and Key Behavioural Drivers

1. Objective & Dataset

The objective is to calculate the baseline churn rate, Customer Lifetime Value (CLV) and monthly recurring revenue (MRR) loss, to analyse churn by contract type, tenure and payment method, and to recommend actions that reduce churn. The raw file contained 605 rows and 13 columns (customer demographics, subscription, billing, support and churn status).

2. Data Cleaning

Issue found	Count	Treatment
Exact duplicate rows (repeated CustomerIDs)	5	Removed; 600 unique customers remain
Region missing	13	Labelled "Unknown"
TotalChargesINR = "unknown" (text)	10	Estimated as Tenure x MonthlyCharges; converted to numeric
MonthlyChargesINR extreme values (5,164 to 13,858)	12	Values were 12x their plan median (annual figure entered as monthly

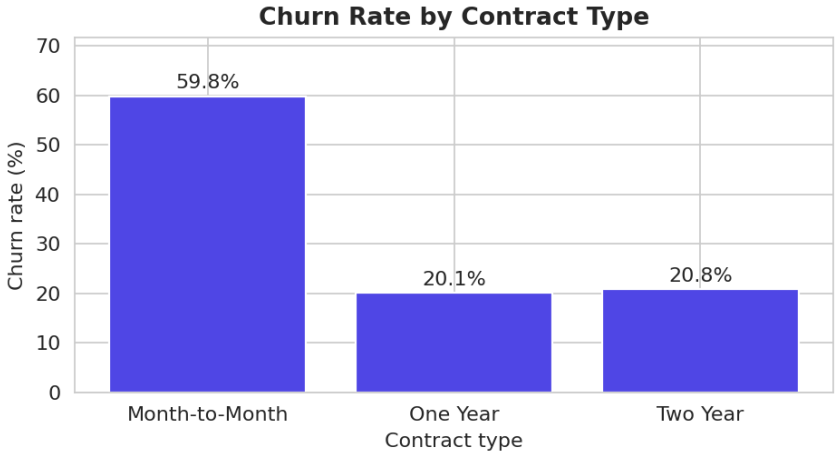
Note: 11 "unknown" TotalCharges were in the raw file; one belonged to a removed duplicate, leaving 10 after de-duplication.

3. Baseline Metrics

Metric	Value	How calculated
Customers analysed	600	After cleaning
Churned customers	247	Churn = Yes
Baseline churn rate	41.2%	247 / 600
Average revenue per customer (ARPU)	INR 759.63 / month	Mean monthly charge
Monthly churn rate	1.16%	Churned customers / total customer-months (21,000+ mont
Expected customer lifetime	~86 months	1 / monthly churn rate
Customer Lifetime Value (CLV)	~INR 65,583	ARPU x expected lifetime
Total MRR (all customers)	INR 4,55,778	Sum of monthly charges
MRR loss from churn	INR 1,87,320 (41.1%)	Sum of monthly charges of churned customers
Avg. revenue collected to date	Churned INR 24,003 vs Retained INR 28,364	Mean TotalCharges

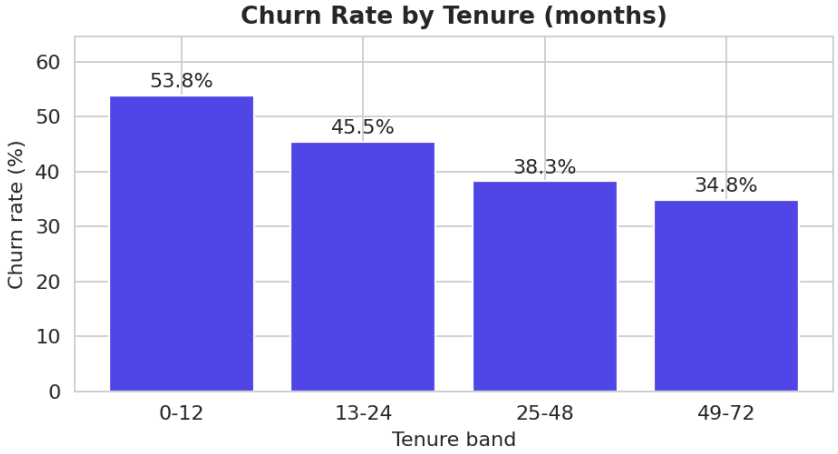
4. Churn by Contract Type, Tenure and Payment Method

Contract	Customers	Churned	Churn rate
Month-to-Month	316	189	59.8%
One Year	164	33	20.1%
Two Year	120	25	20.8%



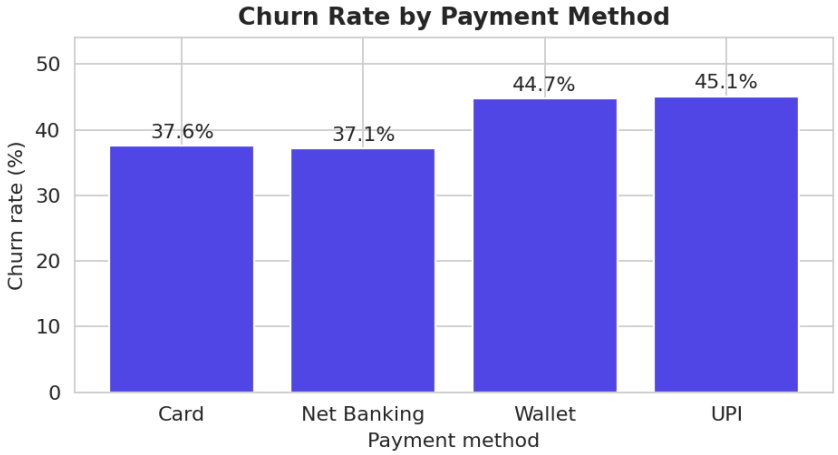
Month-to-month customers churn almost 3x more than annual or two-year customers and account for 189 of the 247 churned customers (77%).

Tenure (months)	Customers	Churned	Churn rate
0-12	104	56	53.8%
13-24	110	50	45.5%
25-48	188	72	38.3%
49-72	198	69	34.8%



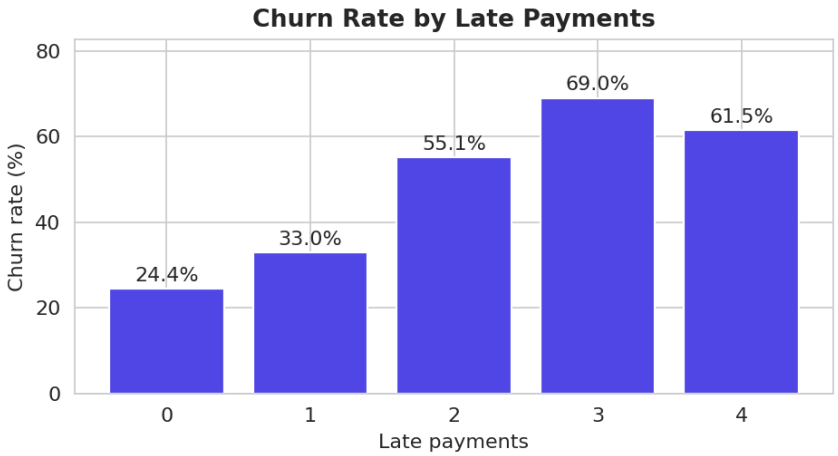
Churn falls steadily with tenure; the first year is the most critical period (53.8%).

Payment method	Customers	Churned	Churn rate
Card	157	59	37.6%
Net Banking	140	52	37.1%
Wallet	161	72	44.7%
UPI	142	64	45.1%



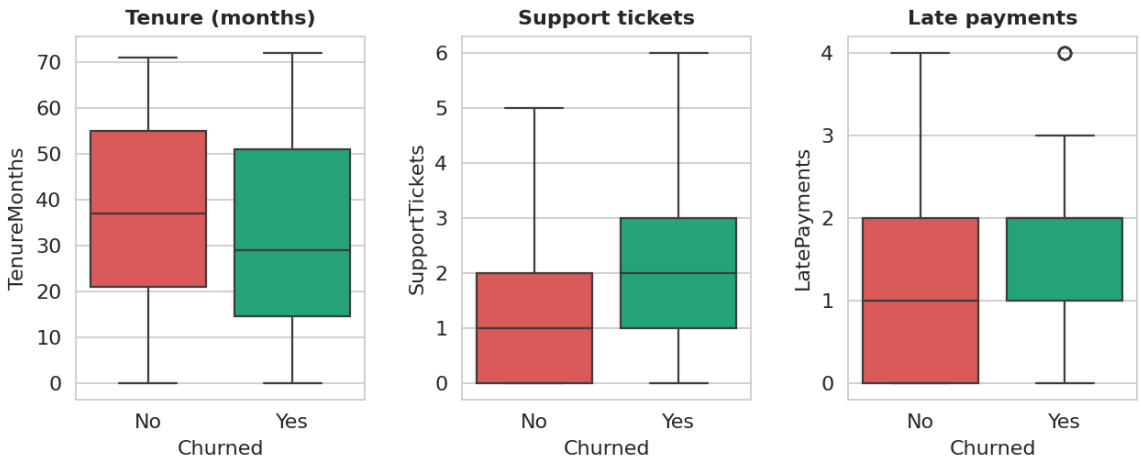
UPI and Wallet users churn about 7-8 percentage points more than Card and Net Banking users. Enrolling in AutoPay, subscription type and age showed no meaningful difference (all about 40-43%), so they are not primary drivers.

5. Behavioural Drivers

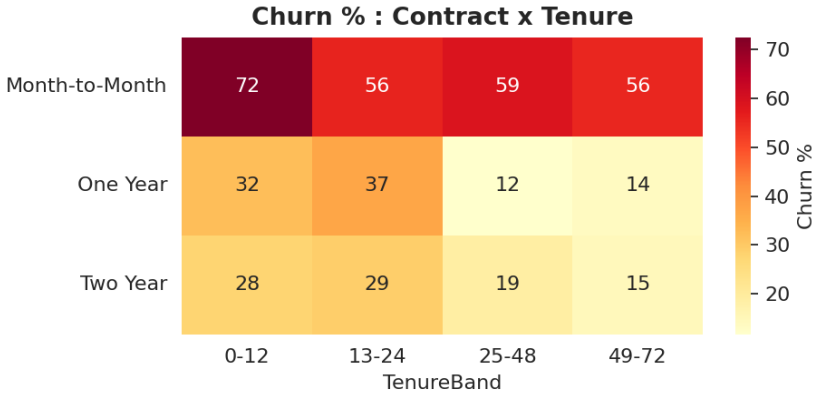


Late payments are the strongest predictor: churn rises from 24.4% (0 late payments) to 55.1% (2) and 69.0% (3). Support tickets show the same pattern: 26.7% churn with 0 tickets versus about 50% with 3 or more (78.6% with 5, small sample).

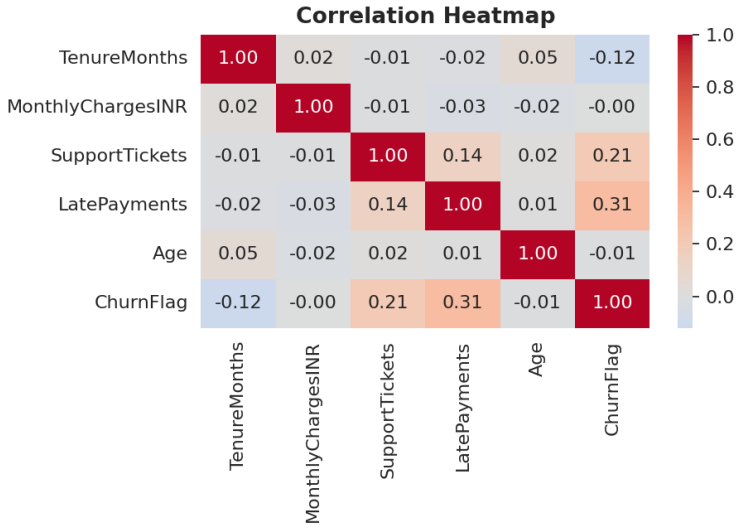
6. Boxplots and Heatmaps



Churned customers have shorter tenure and clearly higher support tickets and late payments than retained customers.



The riskiest segment is month-to-month customers in their first 12 months (72.4% churn). Annual and two-year customers beyond 24 months churn at only 12-19%.



Correlation with churn: late payments +0.31, support tickets +0.21, tenure -0.12; monthly charges and age are about 0.

7. Key Findings

• Baseline churn is 41.2% (247 of 600 customers), causing INR 1.87 lakh (41.1%) of MRR loss; estimated CLV is about INR 65.6K per customer.
• Contract type is the biggest lever: month-to-month churn is 59.8% vs about 20% for longer contracts. These customers alone contribute INR 1.44 lakh of the lost MRR.
• Early-life customers are most at risk: 53.8% churn in the first 12 months, falling to 34.8% after 4 years.
• Late payments and support tickets are strong warning signs (correlation 0.31 and 0.21).
• UPI and Wallet users churn more than Card and Net Banking users.
• Price level, subscription plan, region, age and AutoPay do not differentiate churn materially.

8. Business Recommendations

1. **Convert month-to-month to annual plans:** offer a 10-15% discount or a free month for 12-month commitments. Moving even 20% of this segment could prevent roughly 25 churns.
2. **Strengthen onboarding (first 12 months):** welcome calls, usage check-ins at months 1, 3 and 6, and loyalty rewards at the 12-month mark.
3. **Early-warning system:** flag customers with 2+ late payments or 3+ support tickets and trigger proactive retention outreach (payment reminders, flexible due dates, priority support).
4. **Improve support quality:** reduce repeat tickets via first-contact resolution and follow-ups after ticket closure.
5. **Push stable payment methods:** give small cashback for Card / Net Banking mandates over UPI and Wallet, and send pre-due reminders to reduce failed payments.
6. **Track results:** monitor churn by contract, tenure band and late-payment count monthly; target churn below 35% within two quarters.

Method note: CLV uses the simple formula ARPU x (1 / monthly churn). Associations shown are correlations in this dataset, not proof of causation.